

Mining Social-Network Graphs



Outline

1. Social Networks as Graphs

- a) Examples
- b) Representation
- c) Properties

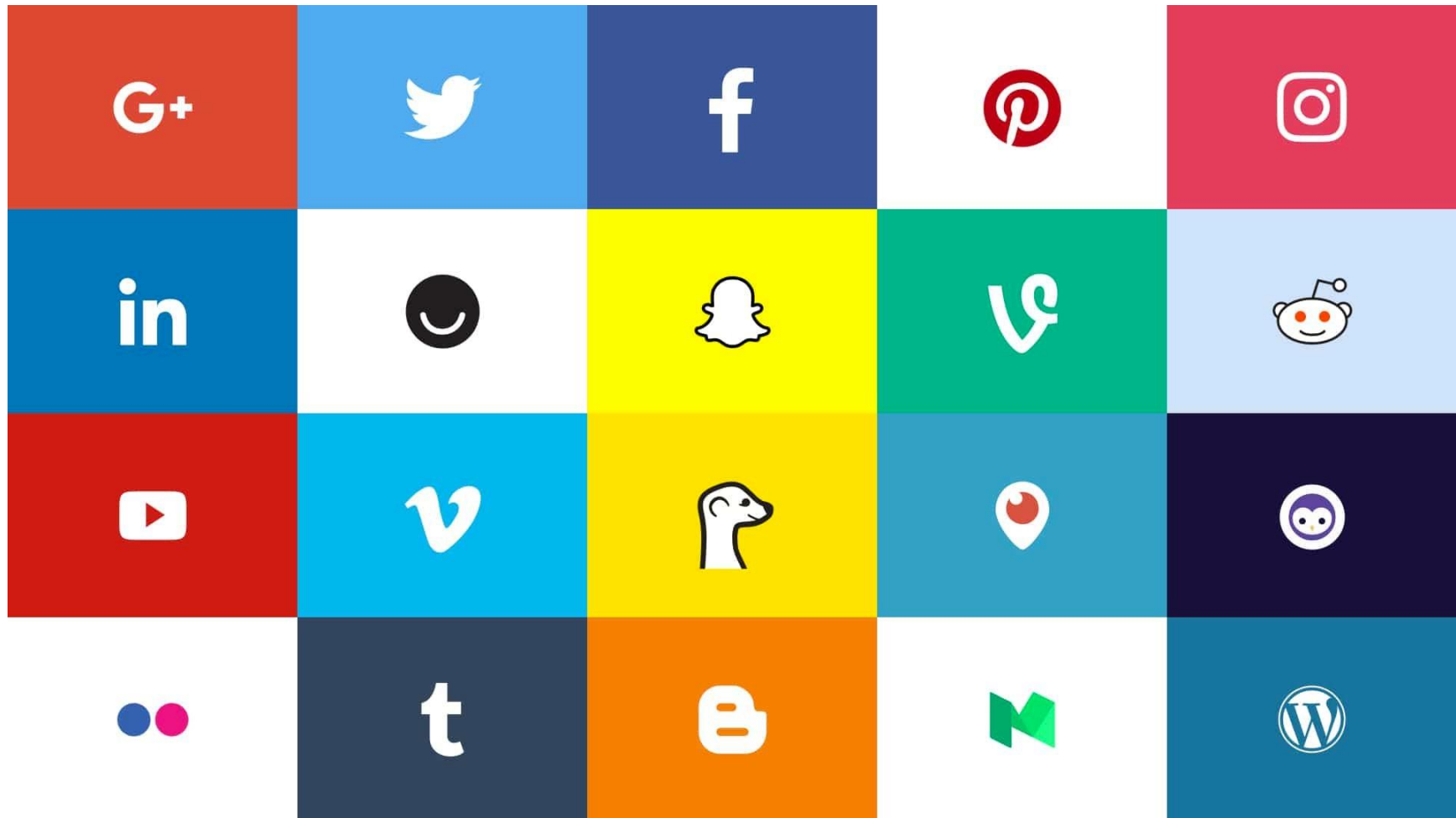
2. Clustering

- a) Distance Measures
- b) Girvan-Newman

3. Spectral Analysis

- a) Networks as Matrices
- b) Connectivity
- c) Partitioning
- d) Clustering

Social Networks (obvious)

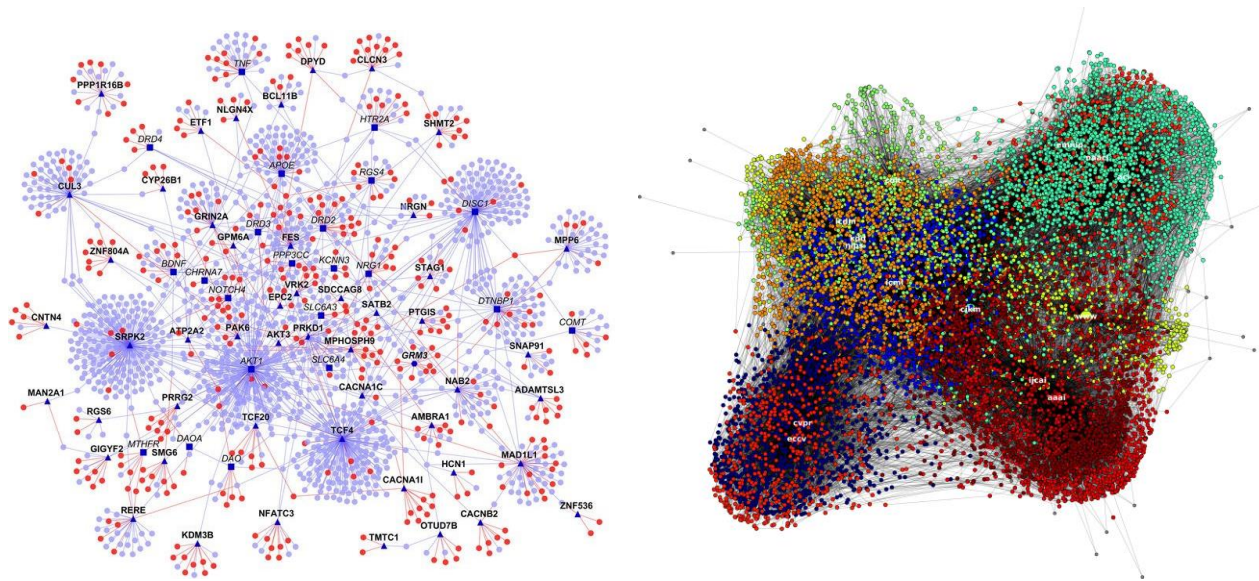


Social Networks (obvious)



Less Obvious

- Phone/E-mail (within K units of time)
- Collaboration (academic papers, patents)
- Biological (proteins, genes)



Representation

- Nodes = entities
 - People, papers, ...
 - **Single** vs. multiple
 - del.icio.us: people, websites, tags
- Edges = relationship
 - **Discrete** vs continuous (i.e. weighted)
 - Directed (e.g. following) vs **undirected**

Checkup

- Given a graph of 7 nodes (A-G), how many undirected edges are possible?

Answer

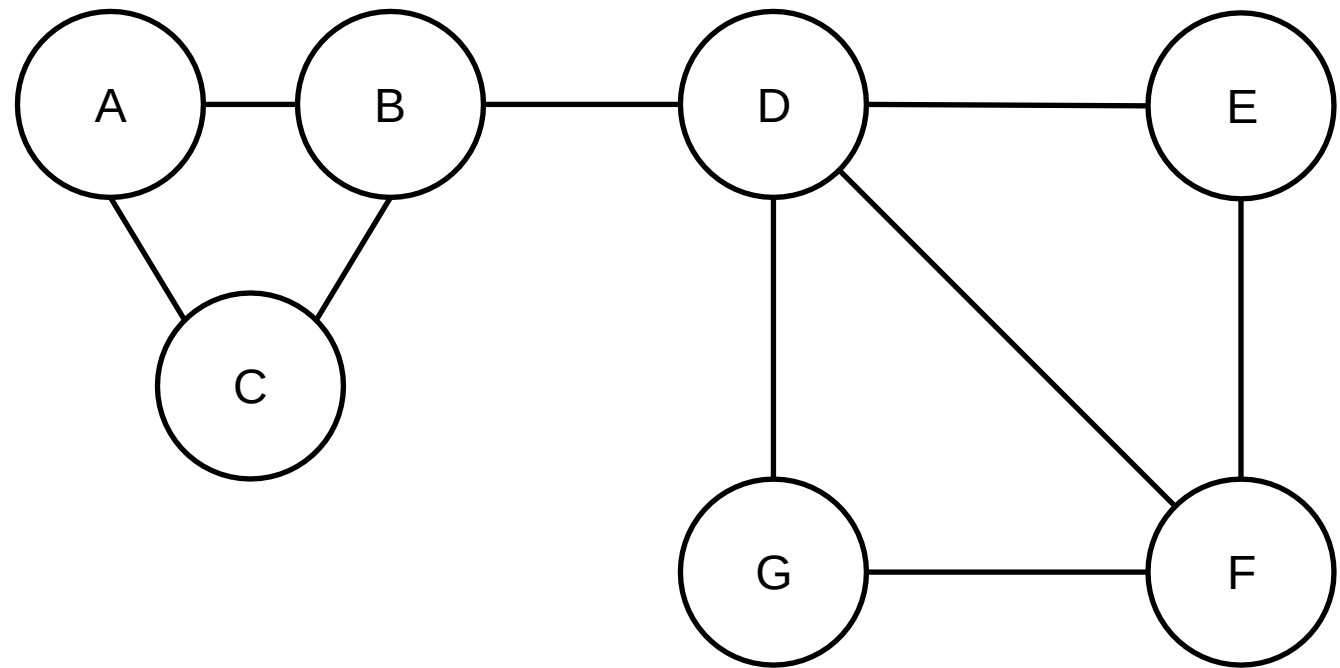
- Given a graph of 7 nodes (A-G), how many undirected edges are possible?

$$\frac{7 \blacklozenge}{2} = \frac{7!}{2!(7-2)!} = \frac{7(7-1)}{2} = 21$$

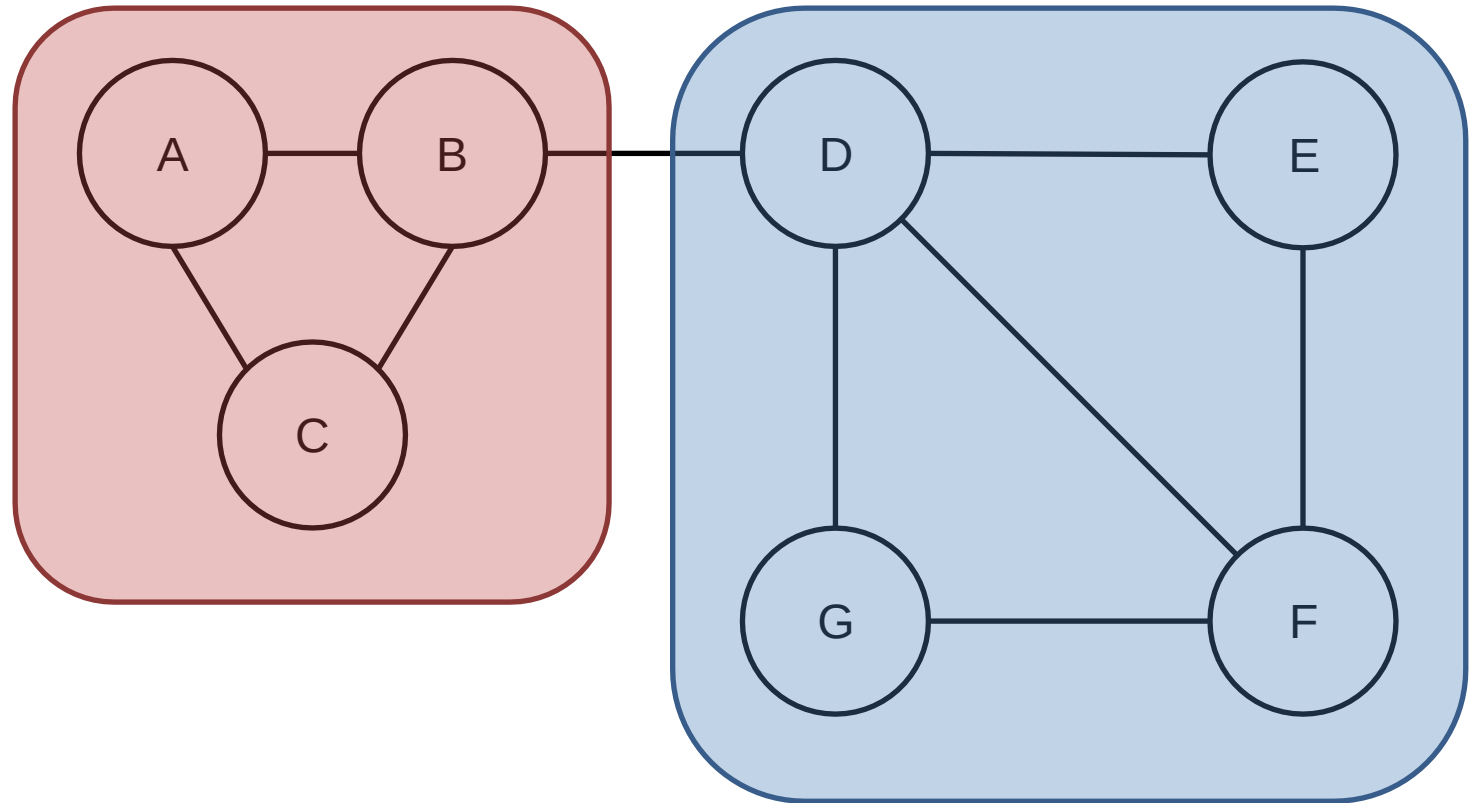
Locality

- Typically assumed that if node A is connected to both B and C, then it is more likely than random that B and C are connected
- Our focus: **Community Detection**
 - Finding groups of densely connected nodes
 - (Tightly related to the locality assumption – i.e. would not work well on random graphs)

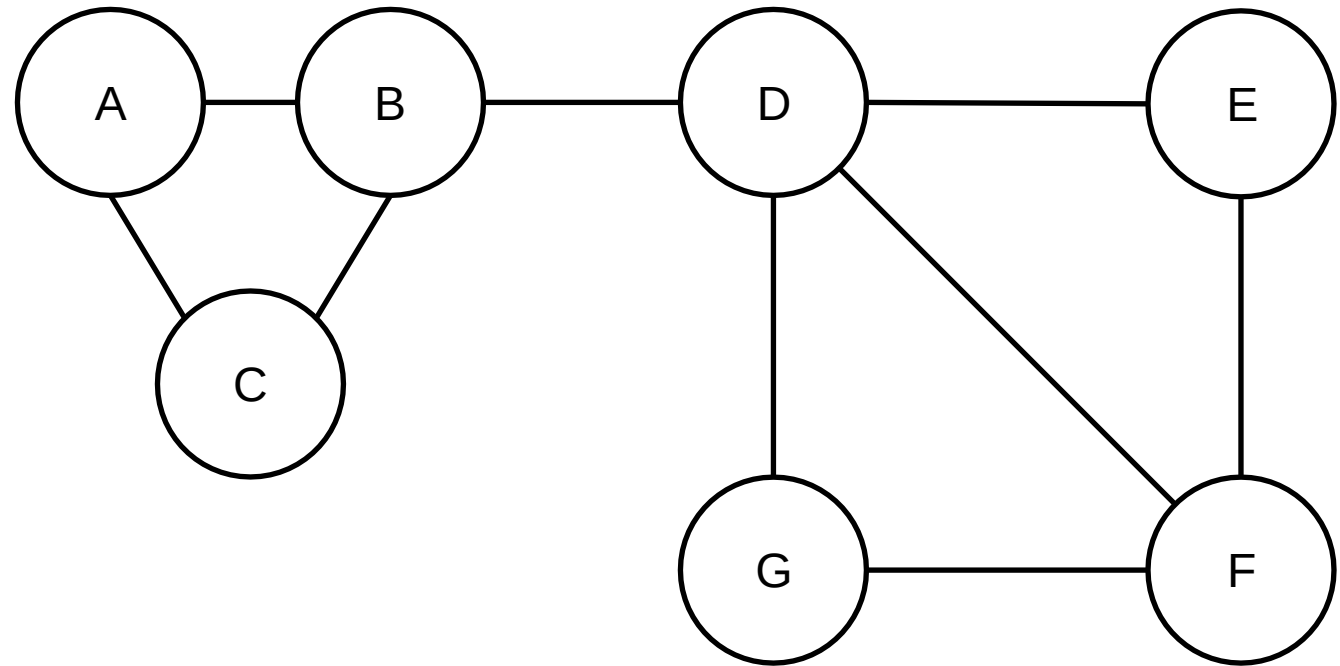
Example Graph



Communities to Be Detected



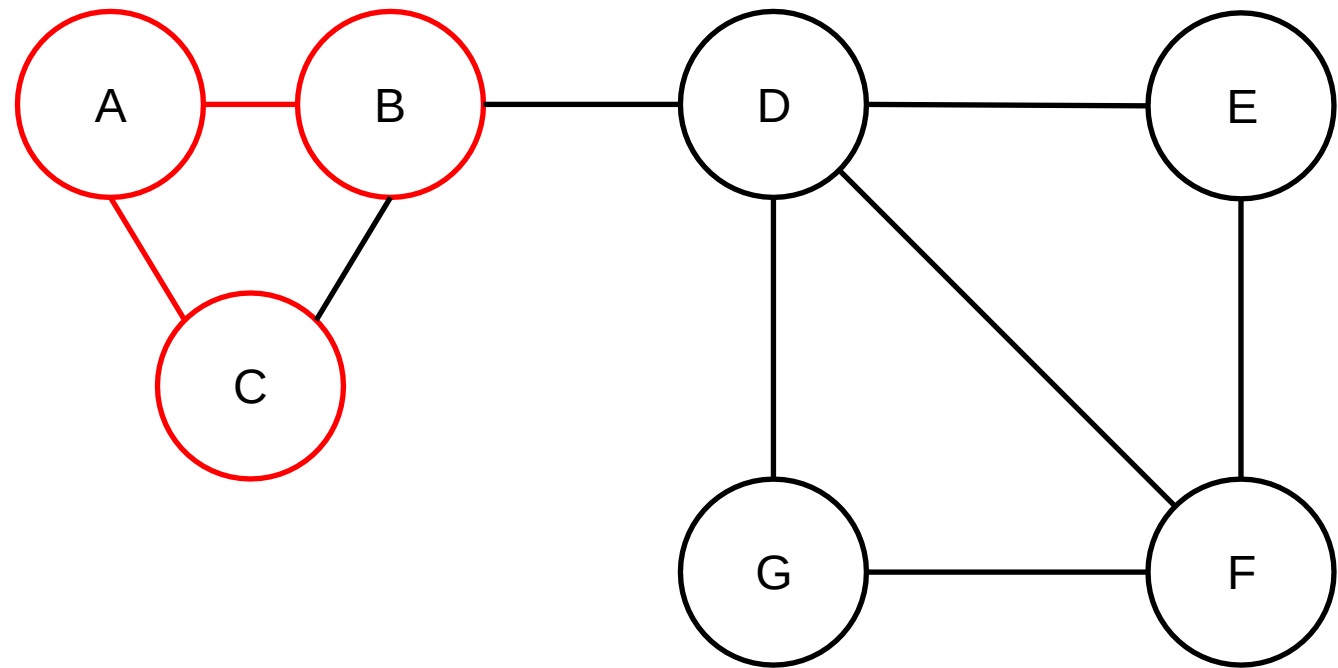
Nodes, Edges, Average Connectivity?



- Nodes: 7
- Edges: 9
- Avg Connectivity: $9/21 \sim 0.43$

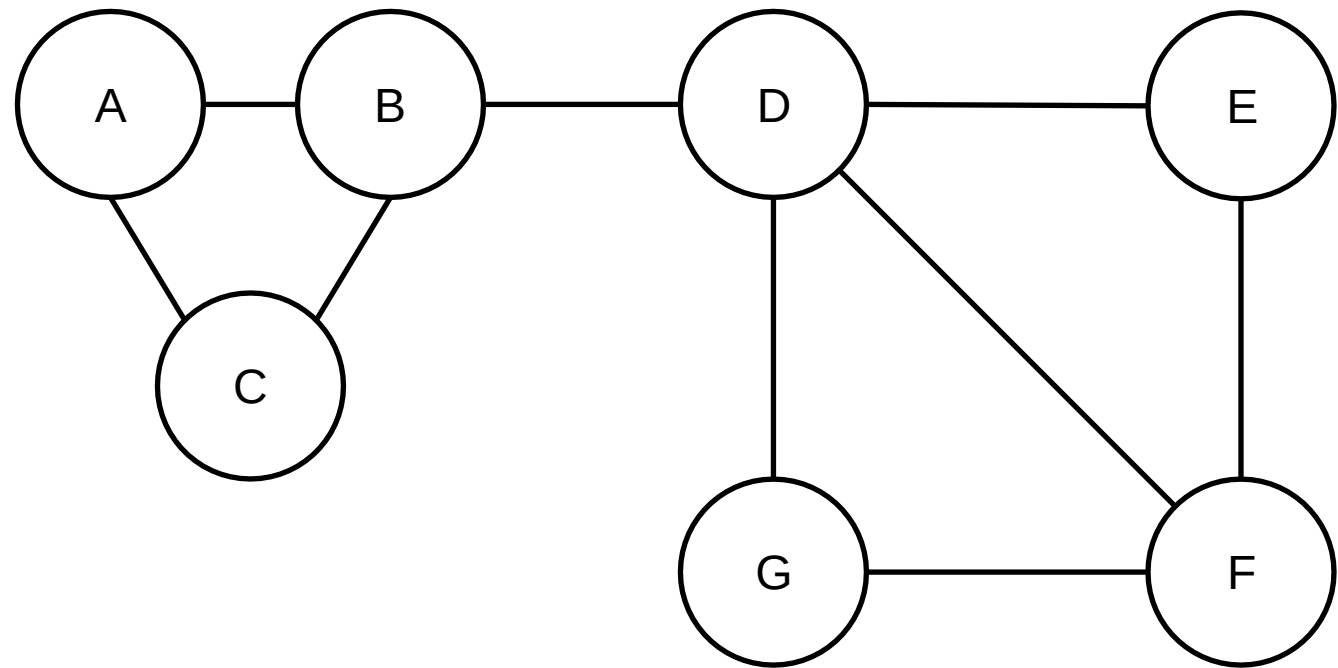
...

Given A-B, A-C; Expected B-C?



- Nodes: 7
- Edges: 9
- Avg Connectivity: $9/21 \sim 0.43$
- Expected Locality: $7/19 \sim 0.37$

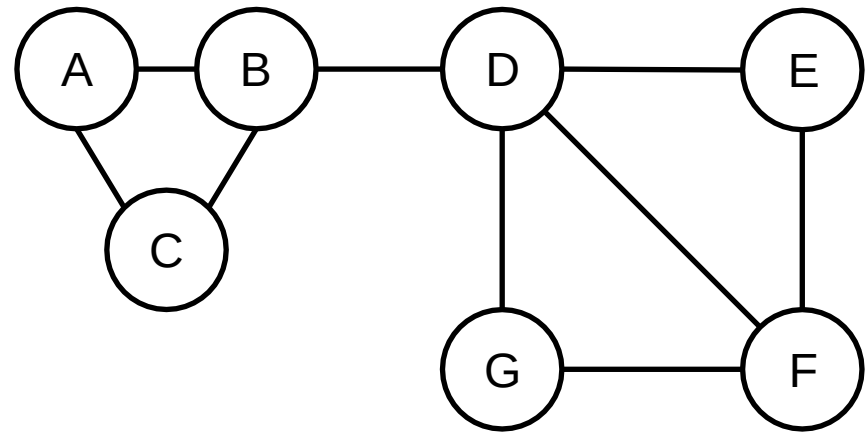
Actual?



- Nodes: 7
- Edges: 9
- Avg Connectivity: $9/21 \sim 0.43$
- Expected Locality: $7/19 \sim 0.37$
- *What are all the triples?*

Evaluate Node Triples

- **A**
 - BC ✓
- **B**
 - AC ✓, AD ✗
 - CD ✗
- **C**
 - AB ✓
- **D**
 - BE ✗, BF ✗, BG ✗
 - EF ✓, EG ✗
 - FG ✓
- **E**
 - DF ✓

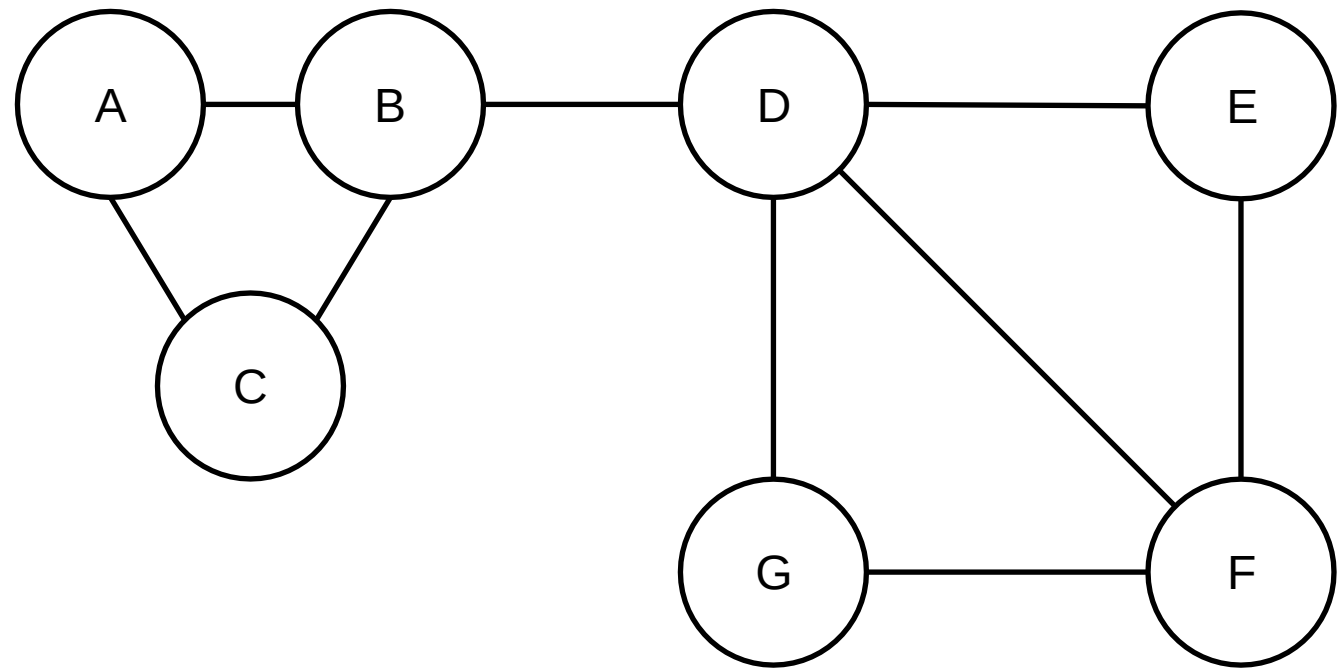


- **F**
 - DE ✓, DG ✓
 - EG ✗
- **G**
 - DF ✓

• ✓ = 9, ✗ = 7

• ✓ / (✓ + ✗) = 9/16 ~ 0.56

Local!

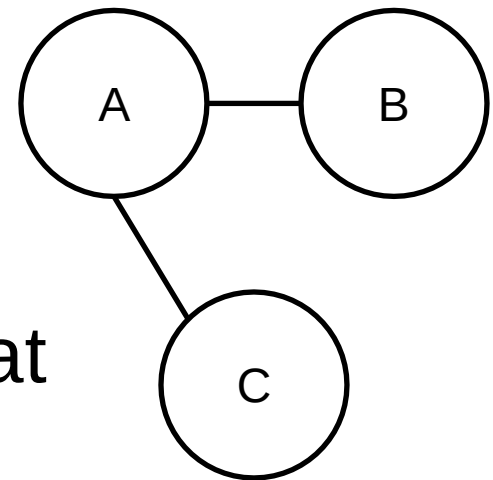


- Nodes: 7
- Edges: 9
- Avg Connectivity: $9/21 \sim 0.43$
- Expected Locality: $7/19 \sim 0.37$
- Actual Locality: $9/16 \sim 0.56$ (>> **Expected!**)

Let's Cluster!

Distance Measure?

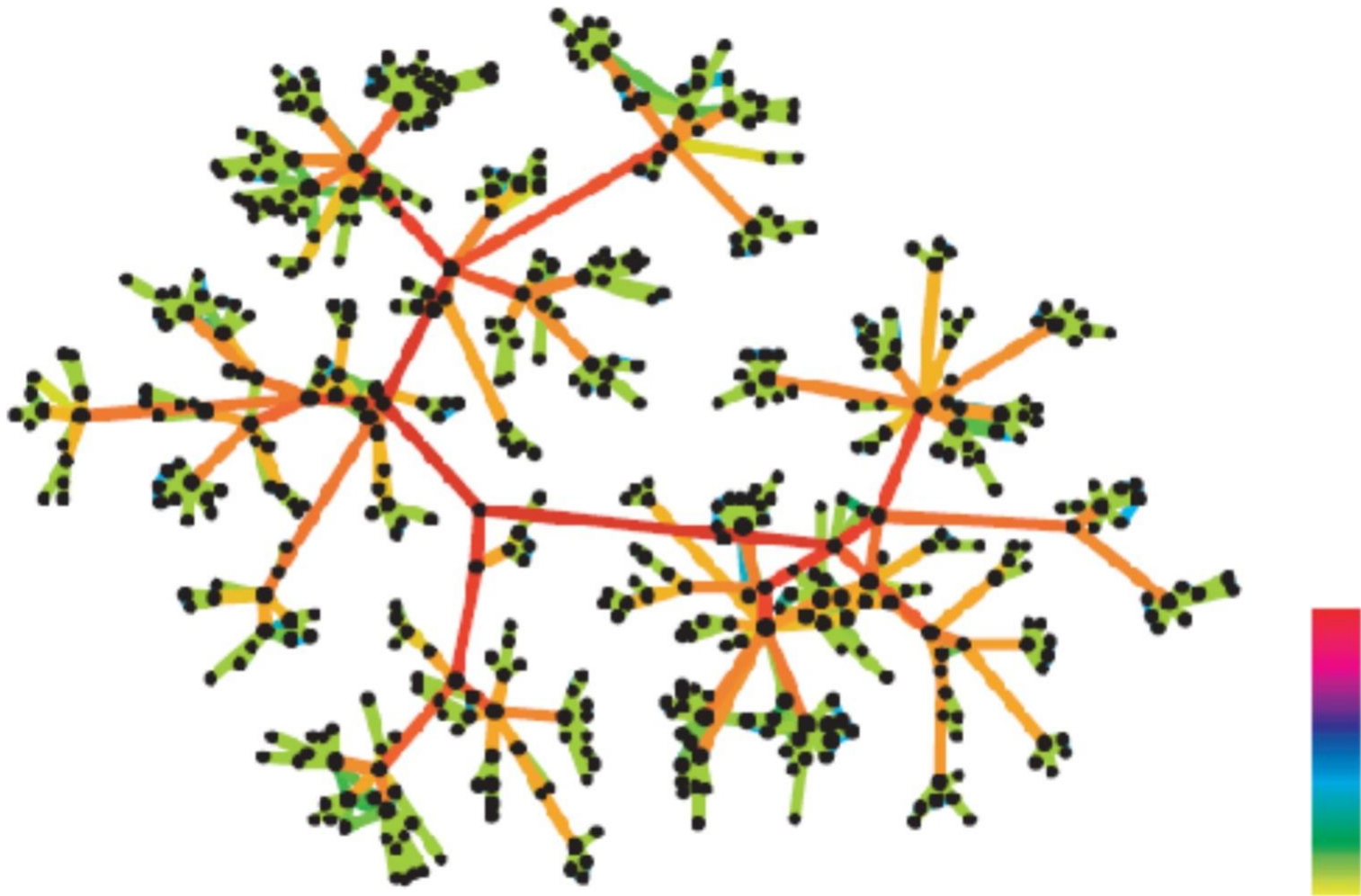
- Distance measures on social-network graphs can be tricky
- KISS ex: $D(x,y)=0$ if edge, 1 otherwise
 - $D(A, B)=0$
 - $D(A, C)=0$
 - $D(B, C) = 1 > D(A, B) + D(A, C)$
- Also leads to unfortunate ties that lead to unfortunate results in typical clustering algorithms



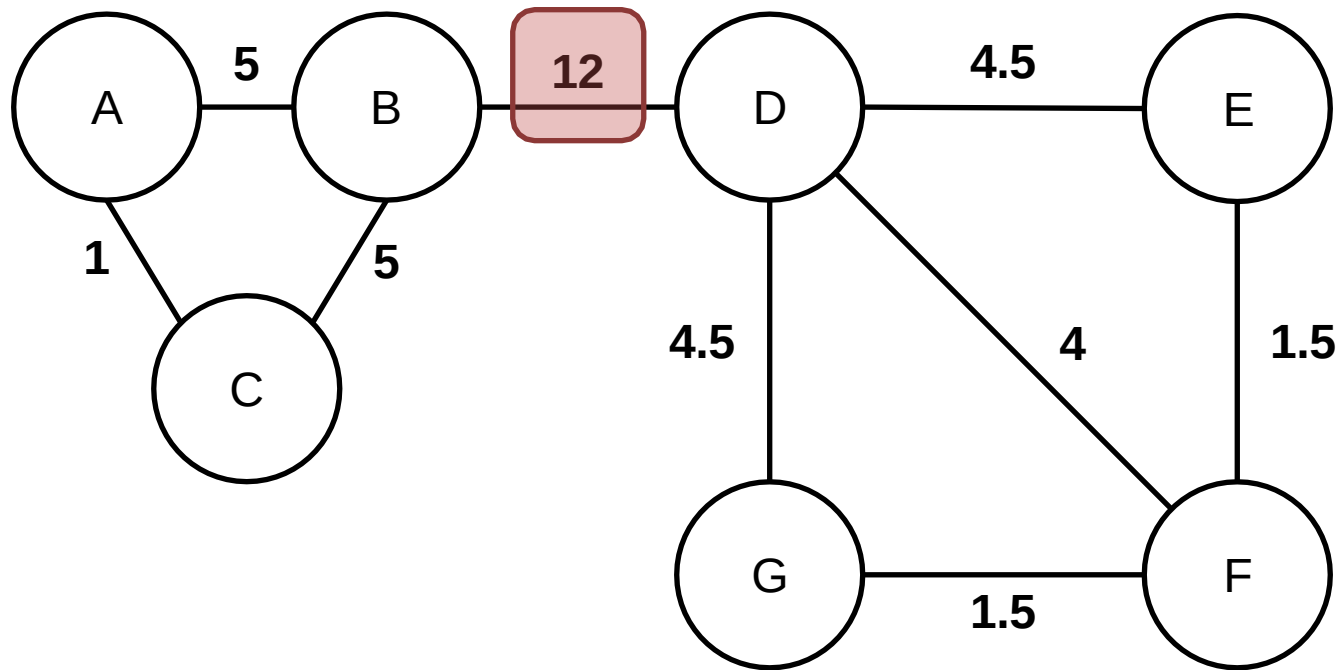
Betweenness

- One of the simplest measures, based on finding the edges that are the *least likely* to be inside a community
 - Clustering: remove high betweenness first!
- $B(a, b)$ = for all pairs of nodes (x, y) , fraction of shortest paths that include edge ab

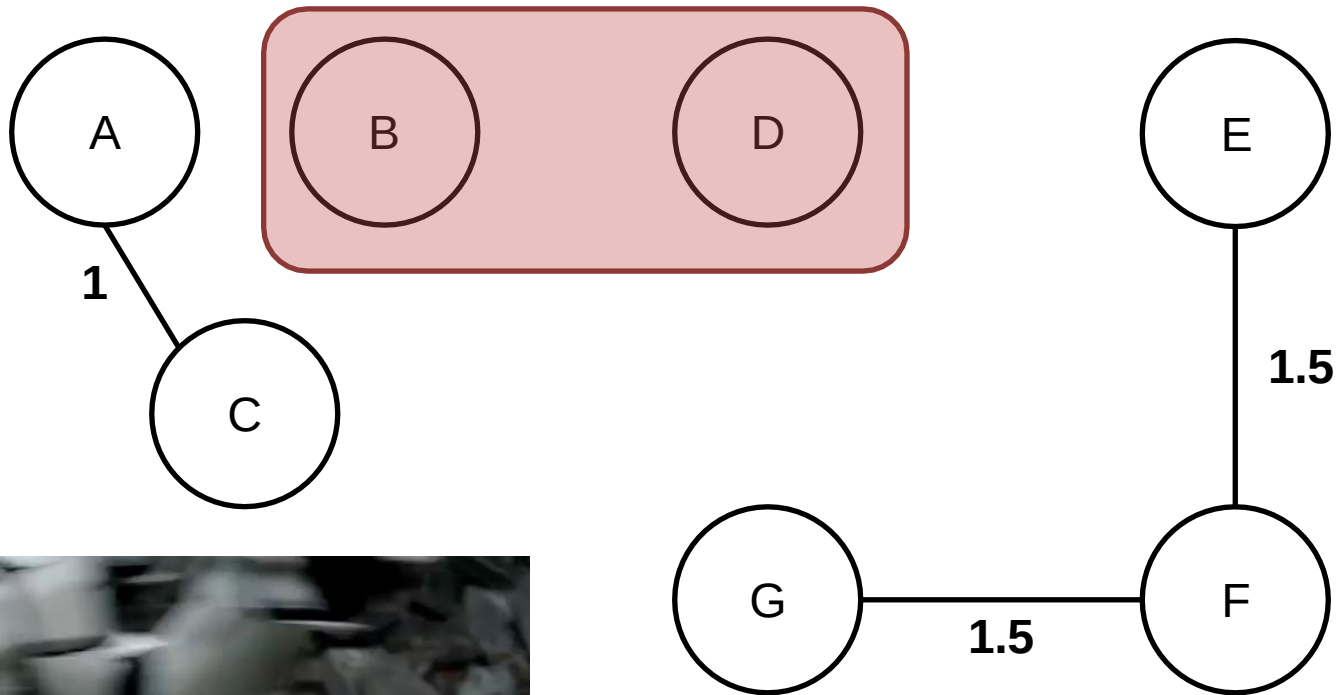
Betweenness Intuition (1)



Betweenness Intuition (2)



Betweenness Intuition (3)



Computing Graph Betweenness

1. For each node

a) Breadth-First Search (BFS)

- WHY?

b) For each edge, compute betweenness

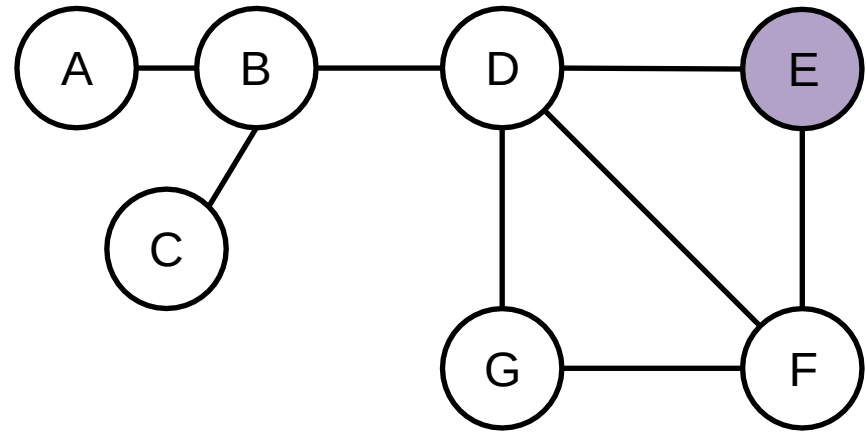
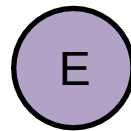
2. For each edge

a) Sum contributions from trees above

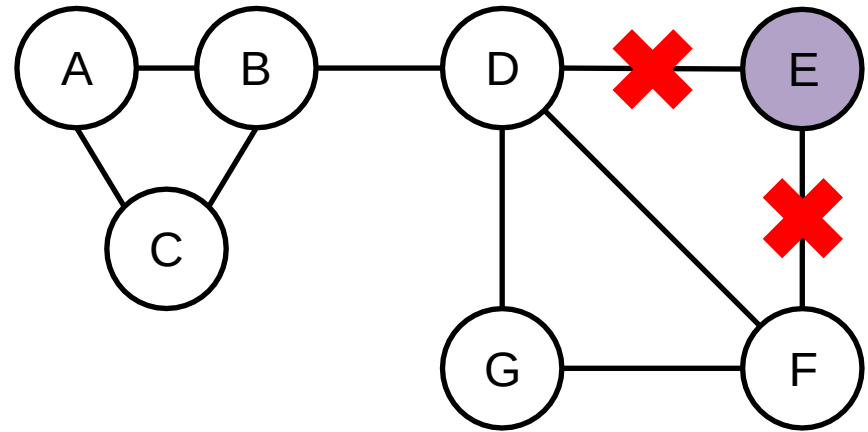
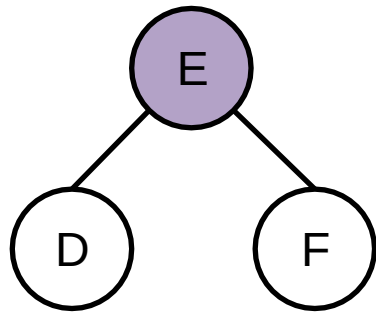
b) Divide by 2 (prevents duplication)

...

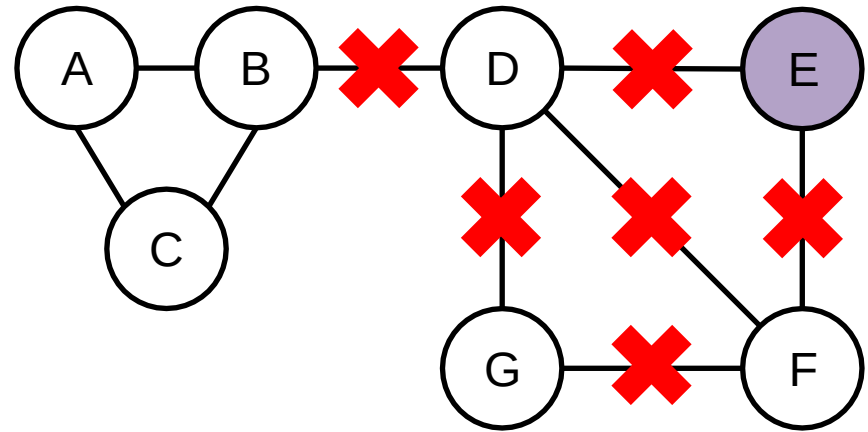
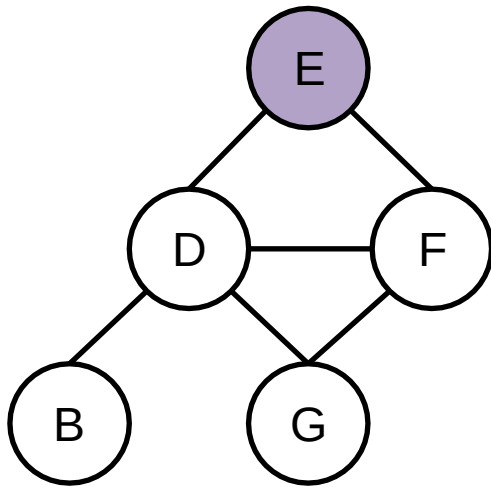
Example (E, BFS.0)



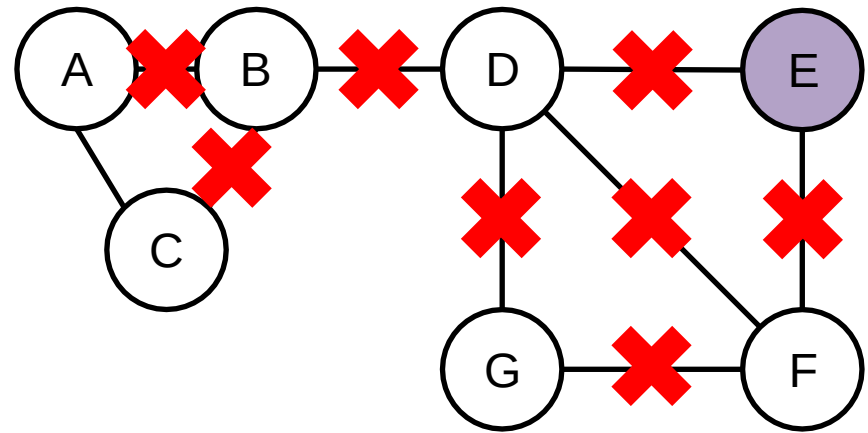
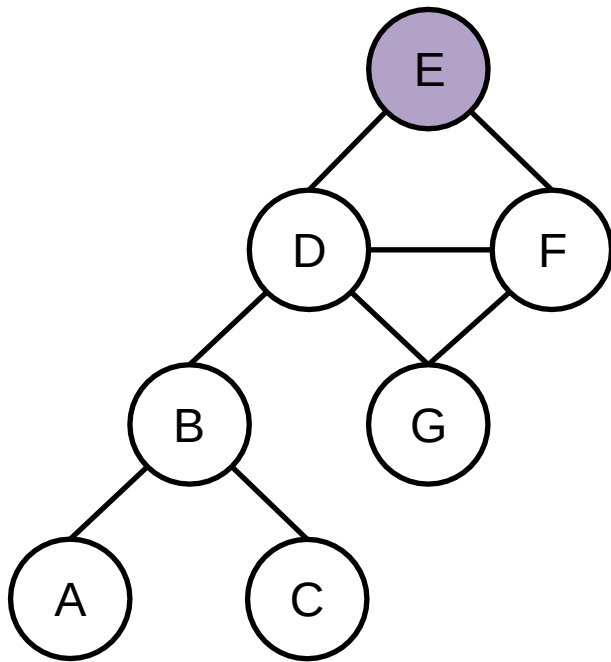
Example (E, BFS.1)



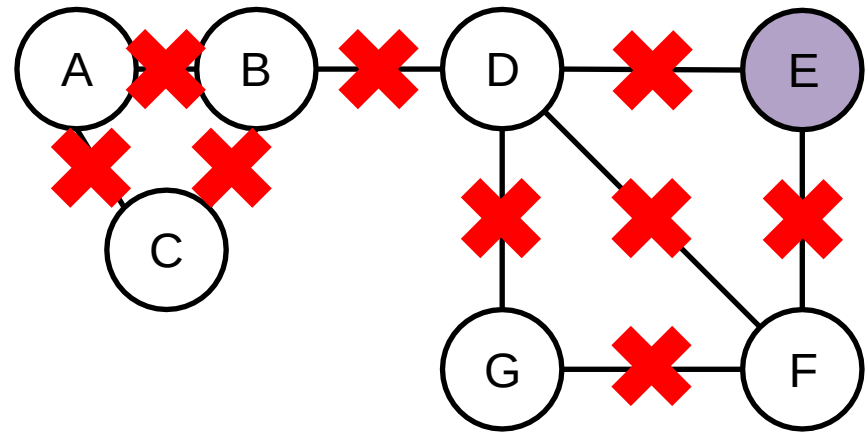
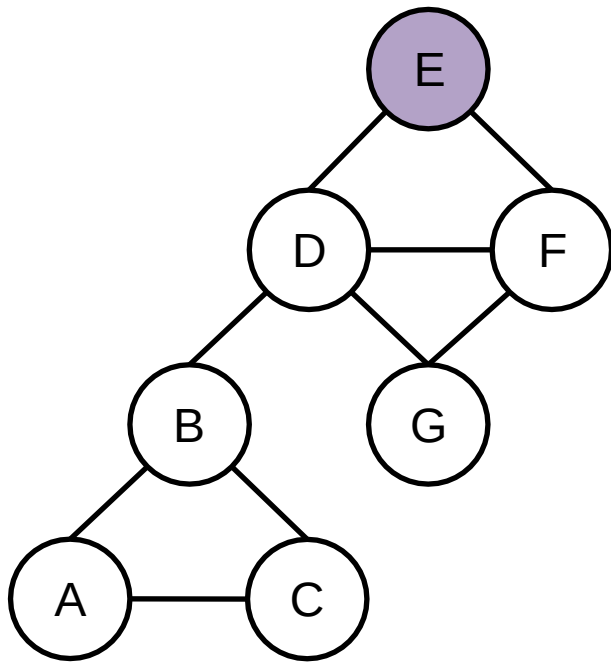
Example (E, BFS.2)



Example (E, BFS.3)



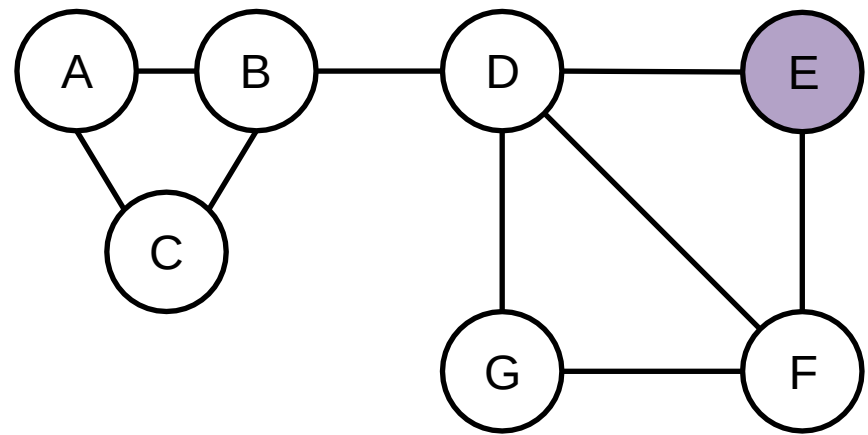
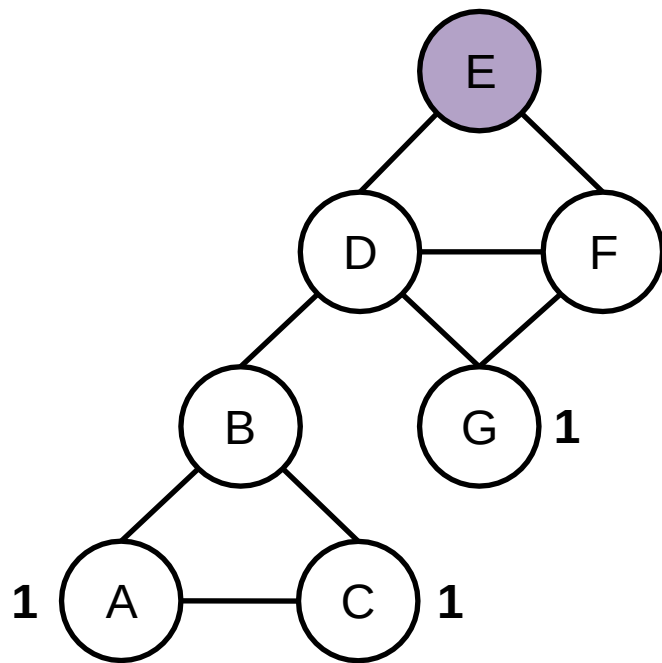
Example (E, BFS.Done)



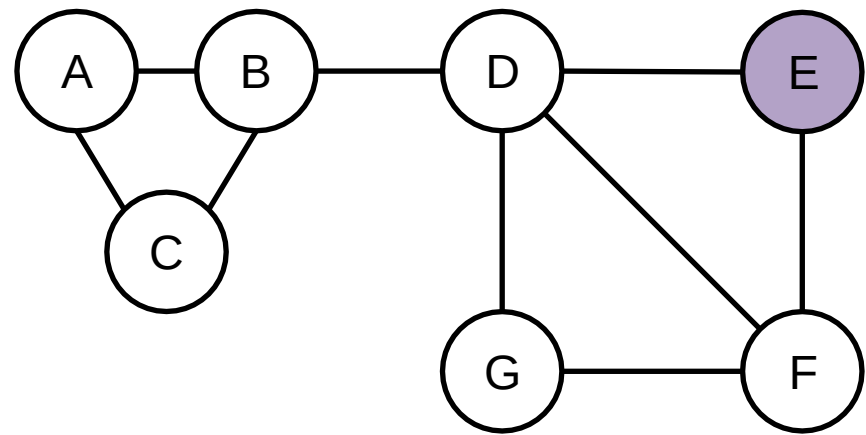
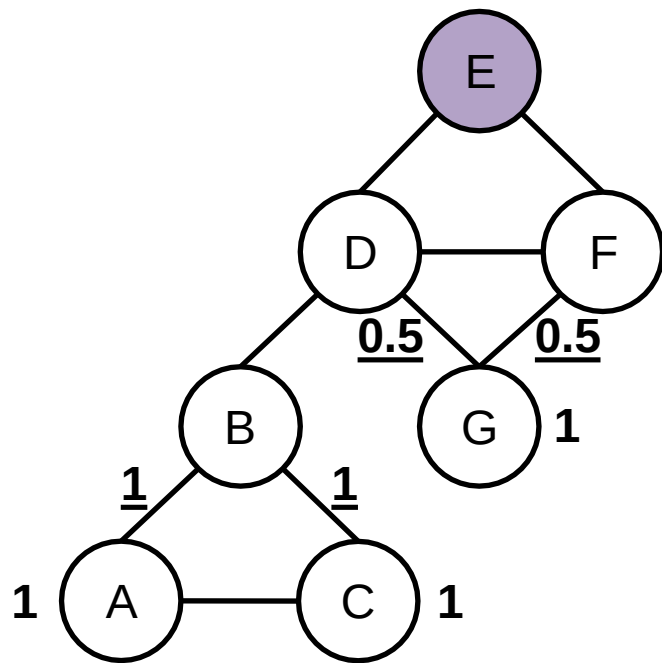
Computing BFS Betweenness

- Recursive credit definition
 - Node gets $1 + \text{sum of edge credits below}$
 - Edge gets node below divided by outgoing
 - Ignore between nodes at the same level
(never used for shortest paths)
- Start from leaves, move up

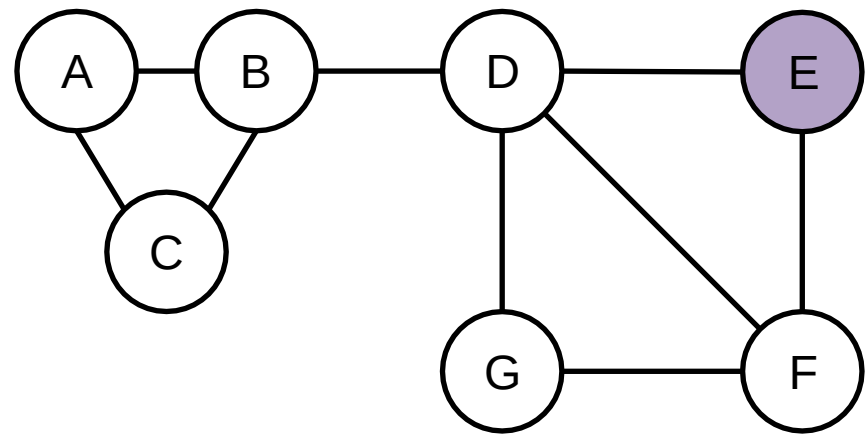
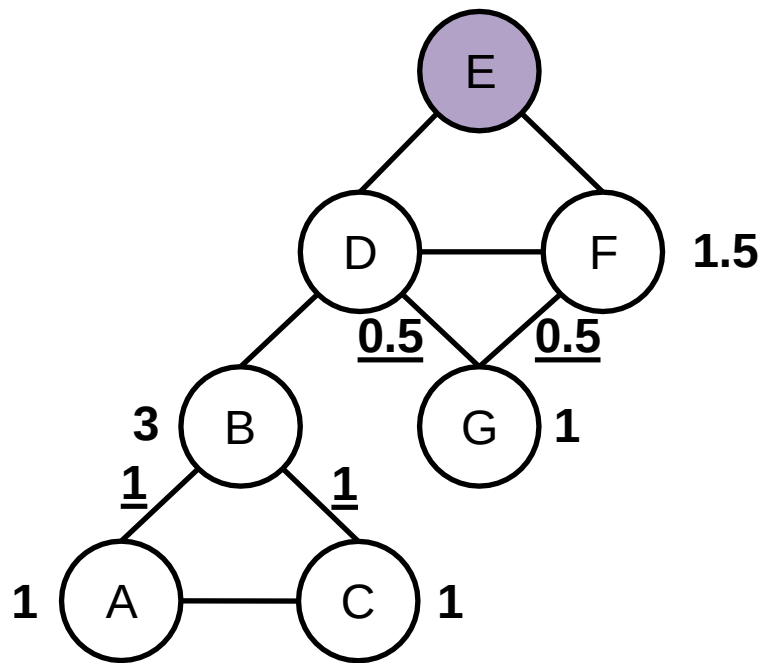
Example (E, Credits.1)



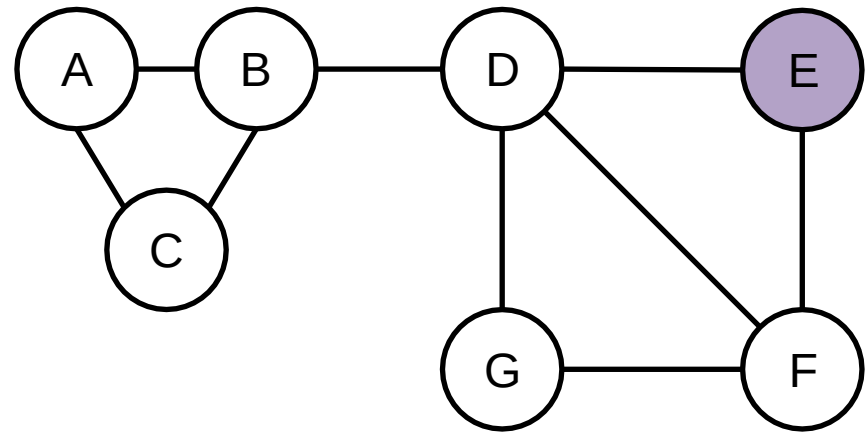
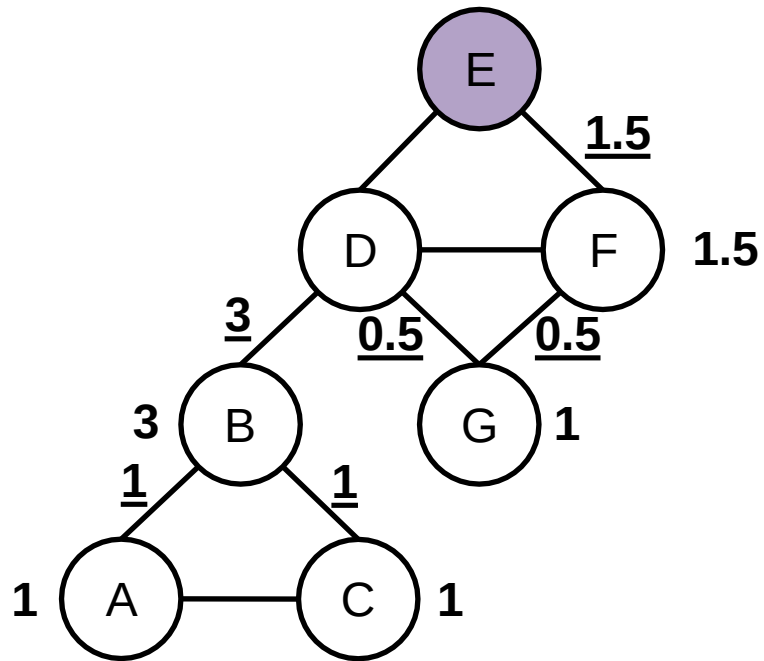
Example (E, Credits.2)



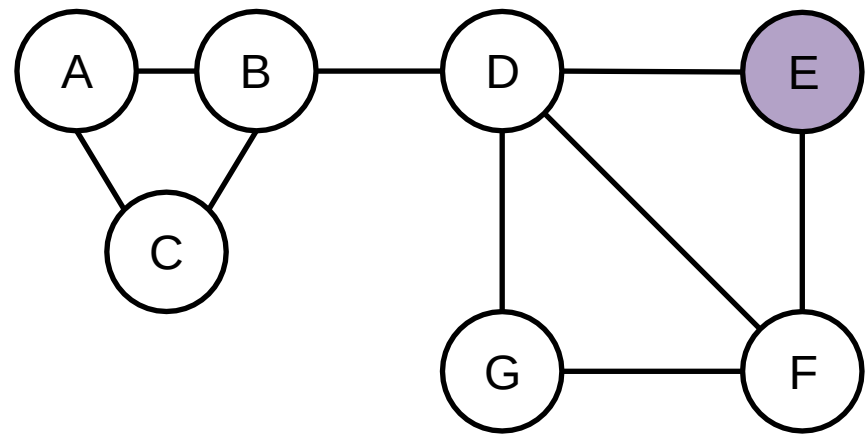
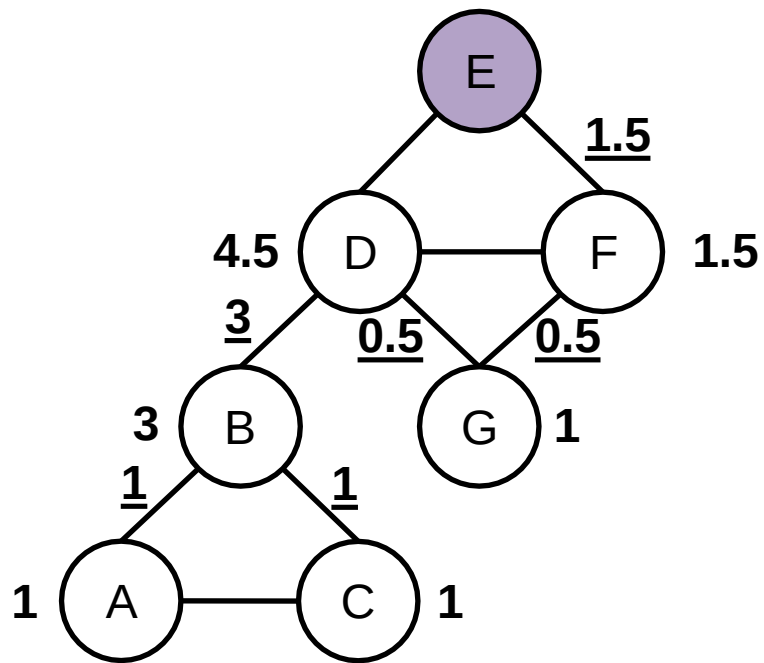
Example (E, Credits.3)



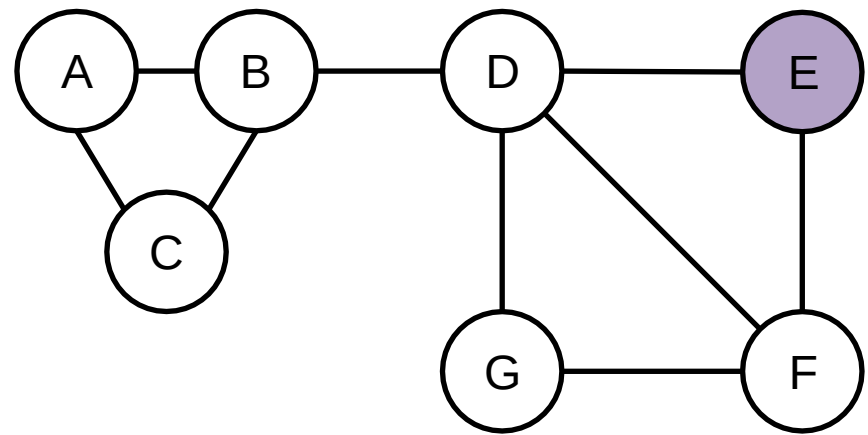
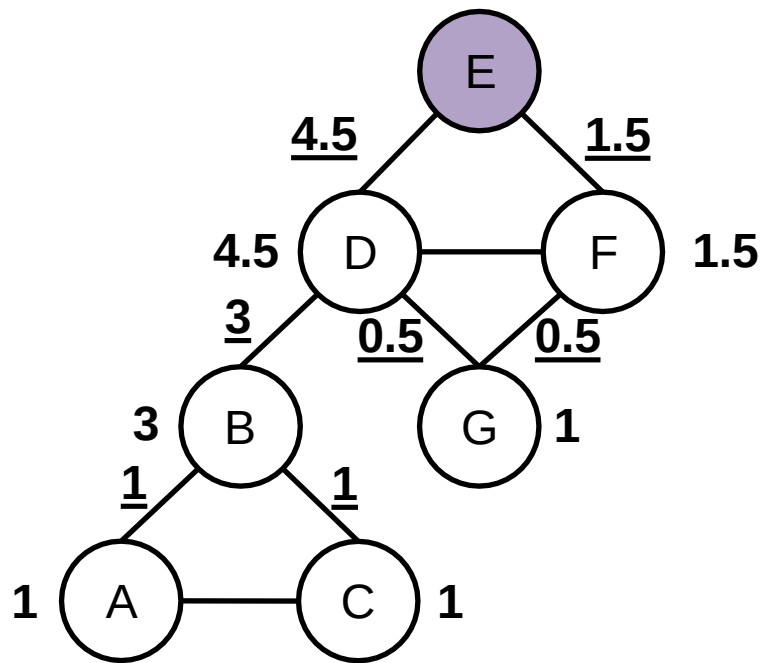
Example (E, Credits.4)



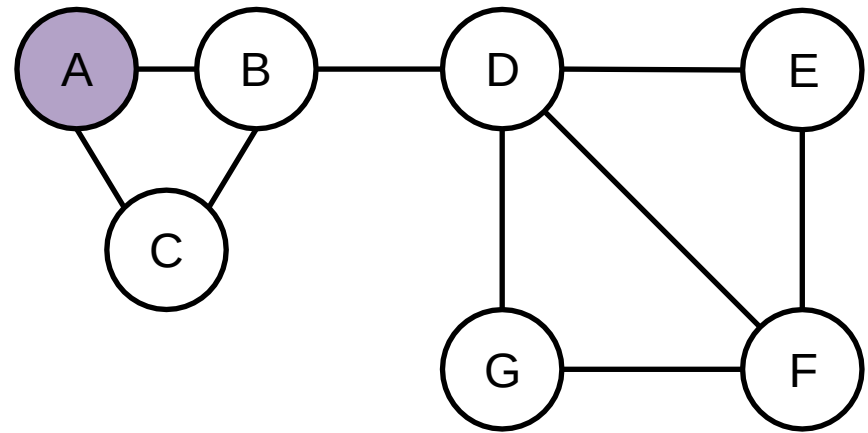
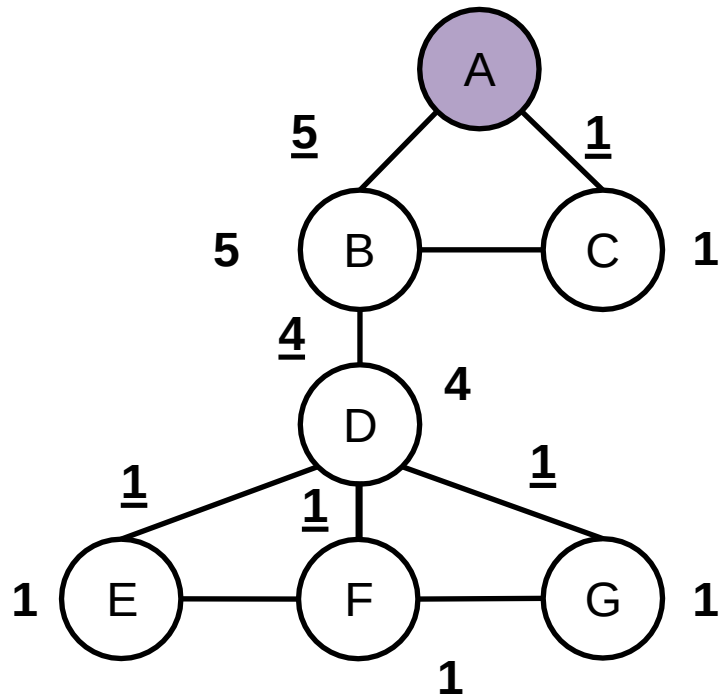
Example (E, Credits.5)



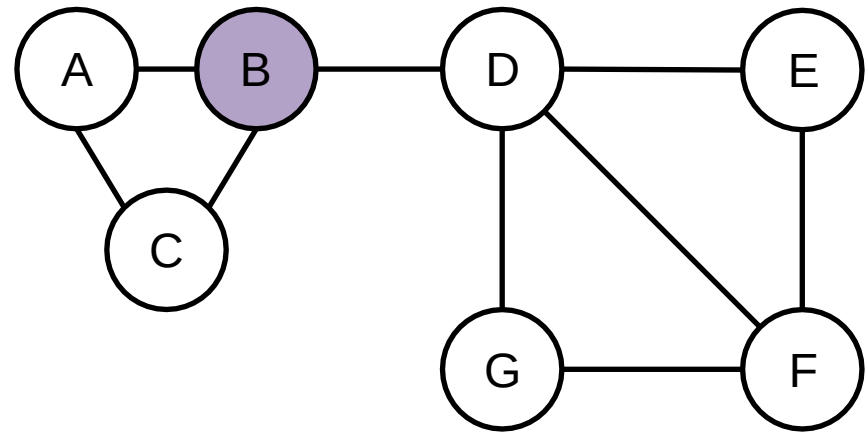
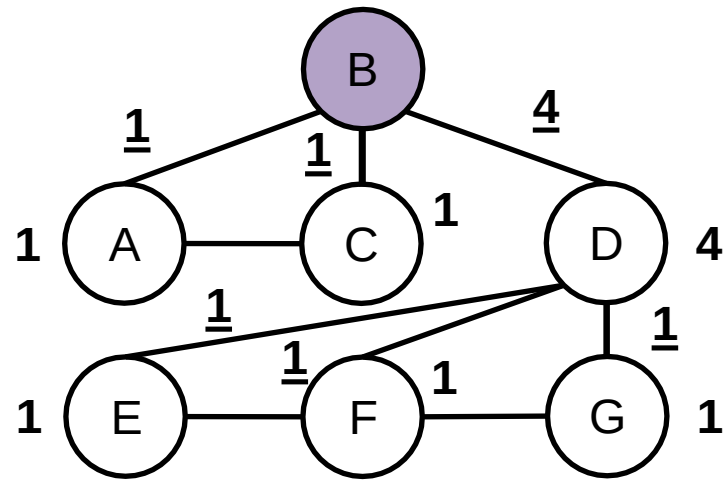
Example (E, Credits.6)



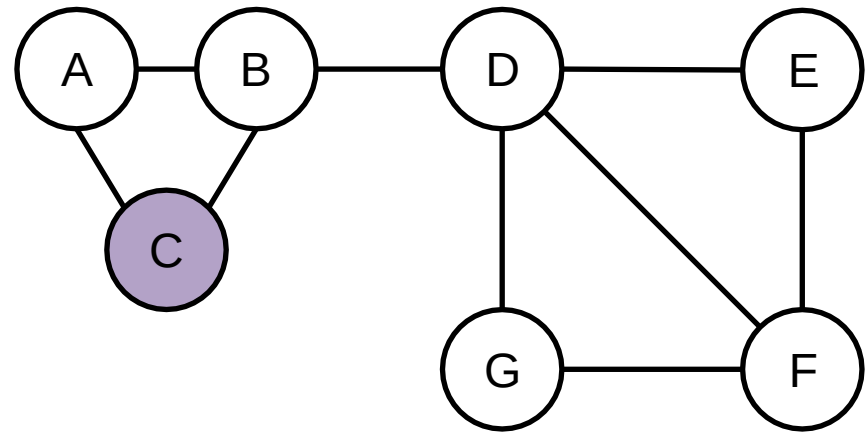
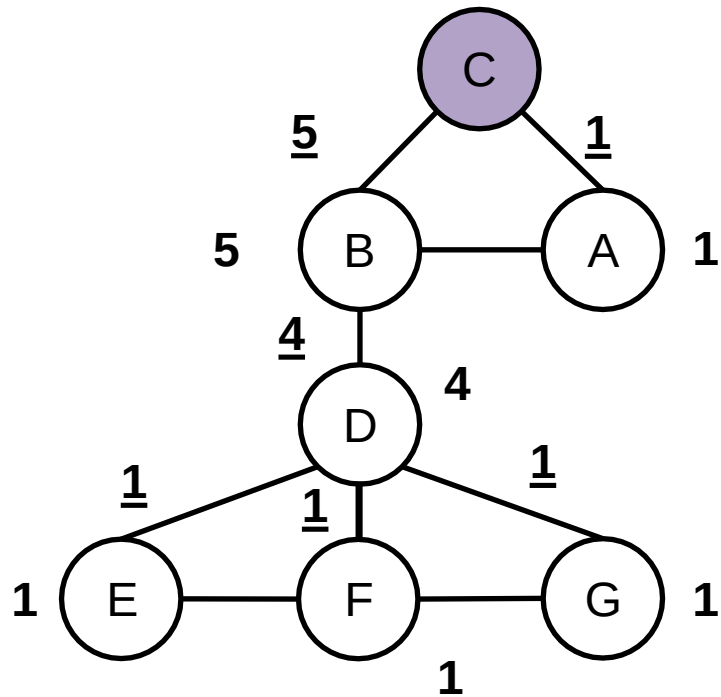
Example (A, Credits)



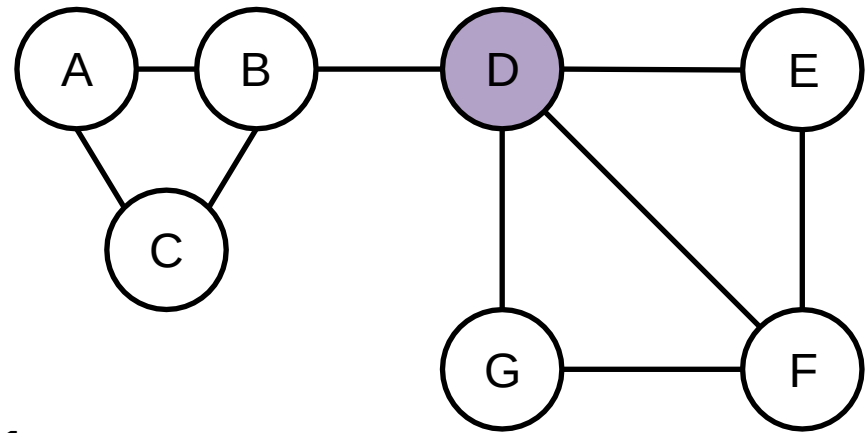
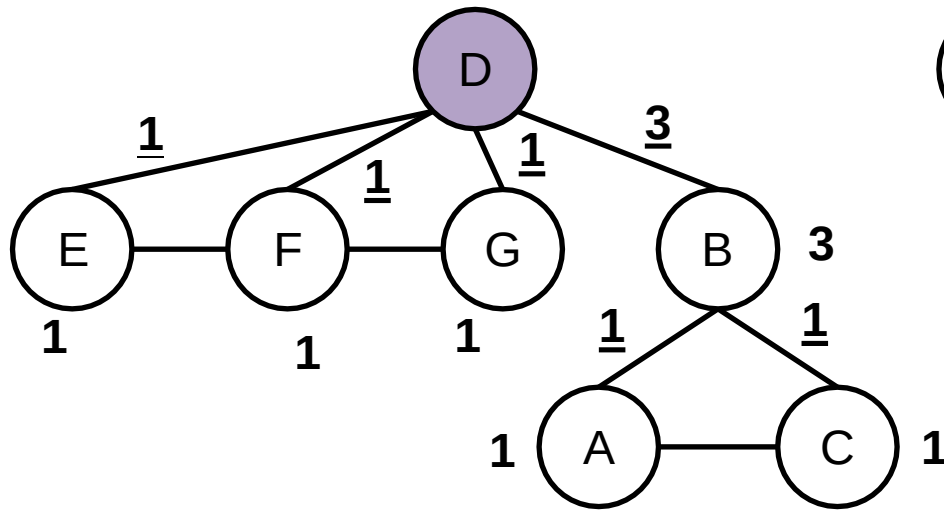
Example (B, Credits)



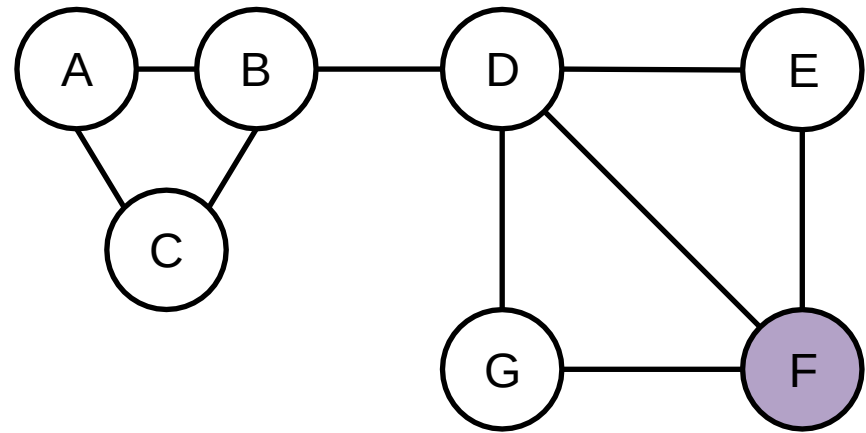
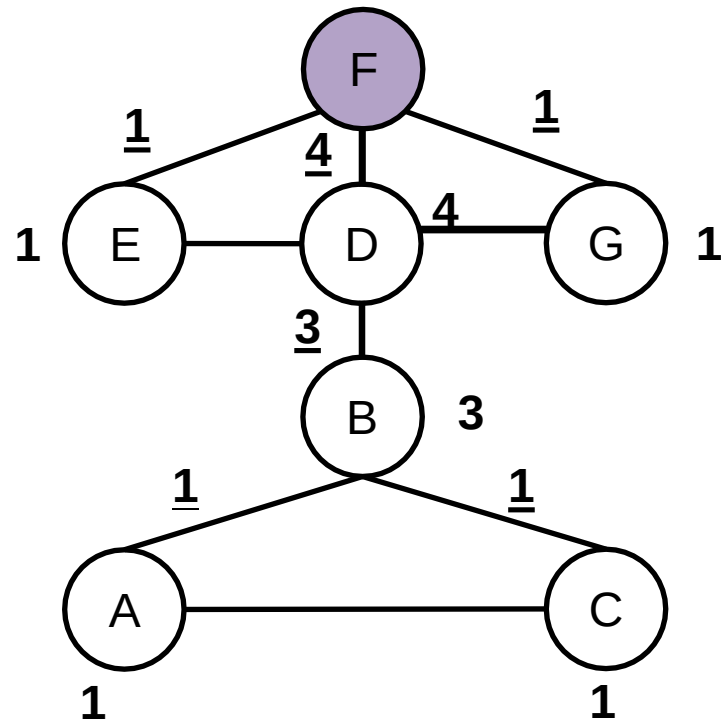
Example (C, Credits)



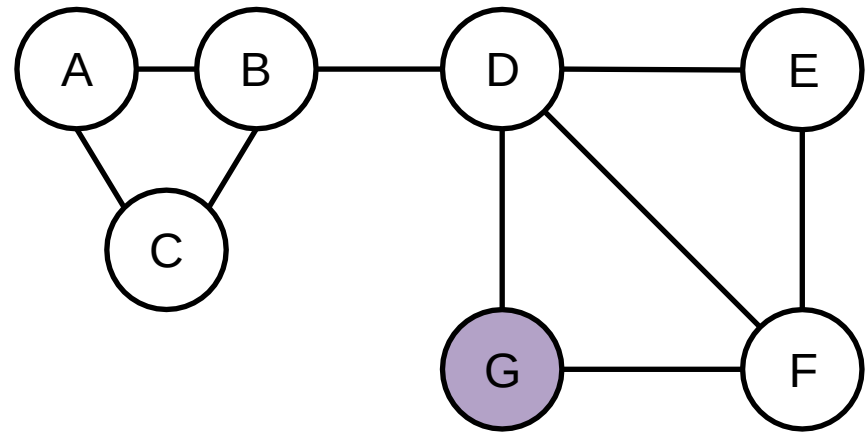
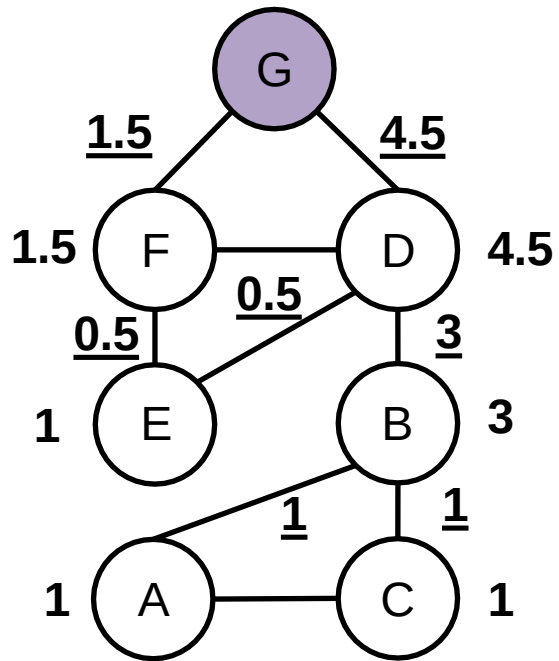
Example (D, Credits)



Example (F, Credits)

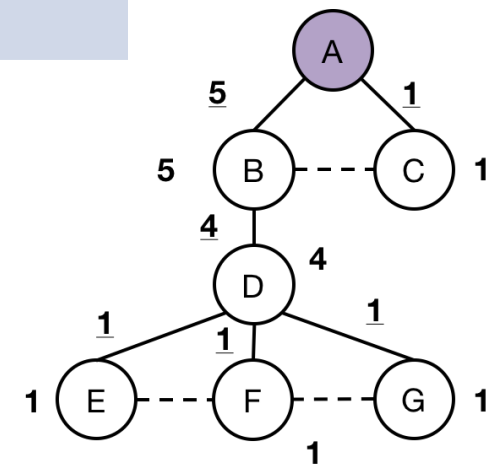


Example (G, Credits)



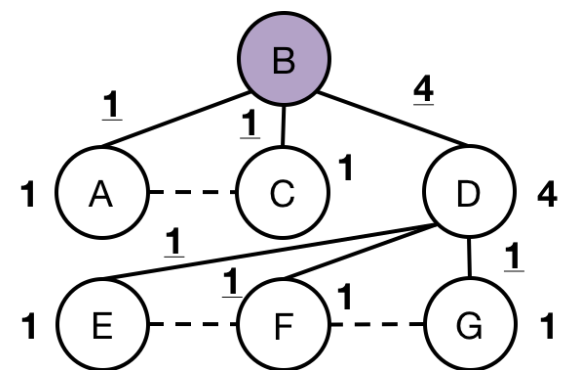
Sum Contributions (A)

	AB	AC	BC	BD	DE	DG	DF	EF	GF
A	5	1		4	1	1	1		
B									
C									
D									
E									
F									
G									



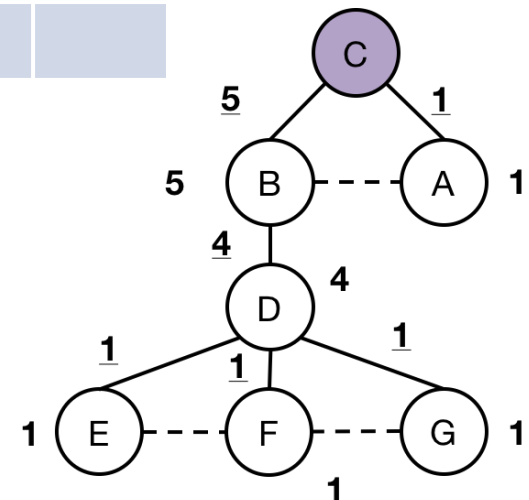
Sum Contributions (B)

	AB	AC	BC	BD	DE	DG	DF	EF	GF
A	5	1		4	1	1	1		
B	1		1	4	1	1	1		
C									
D									
E									
F									
G									



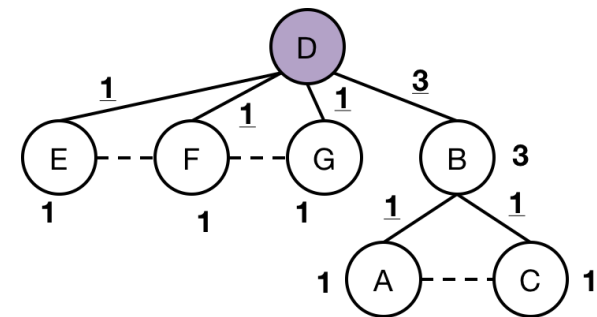
Sum Contributions (C)

	AB	AC	BC	BD	DE	DG	DF	EF	GF
A	5	1		4	1	1	1		
B	1		1	4	1	1	1		
C		1	5	4	1	1	1		
D									
E									
F									
G									



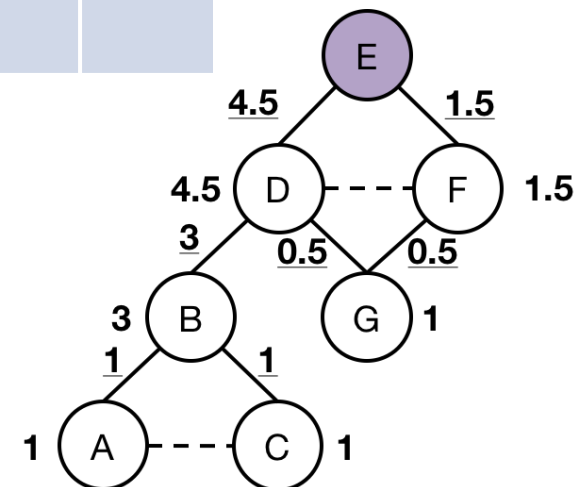
Sum Contributions (D)

	AB	AC	BC	BD	DE	DG	DF	EF	GF
A	5	1		4	1	1	1		
B	1		1	4	1	1	1		
C		1	5	4	1	1	1		
D	1		1	3	1	1	1		
E									
F									
G									



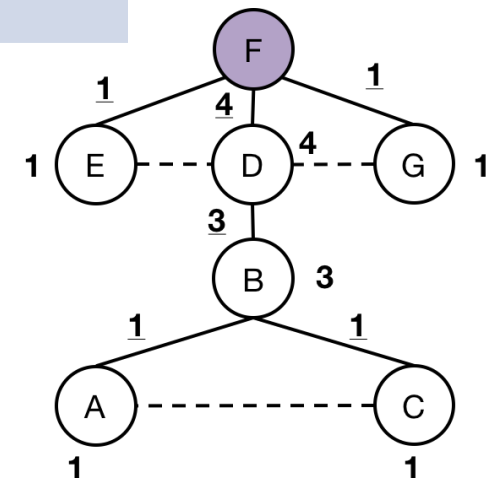
Sum Contributions (E)

	AB	AC	BC	BD	DE	DG	DF	EF	GF
A	5	1		4	1	1	1		
B	1		1	4	1	1	1		
C		1	5	4	1	1	1		
D	1		1	3	1	1	1		
E	1		1	3	4.5	0.5		1.5	0.5
F									
G									



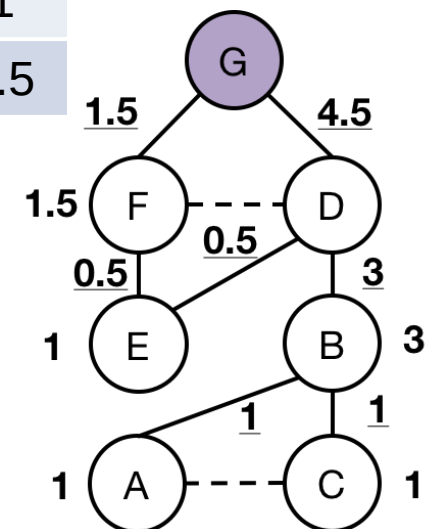
Sum Contributions (F)

	AB	AC	BC	BD	DE	DG	DF	EF	GF
A	5	1		4	1	1	1		
B	1		1	4	1	1	1		
C		1	5	4	1	1	1		
D	1		1	3	1	1	1		
E	1		1	3	4.5	0.5		1.5	0.5
F	1		1	3			4	1	1
G									



Sum Contributions (G)

	AB	AC	BC	BD	DE	DG	DF	EF	GF
A	5	1		4	1	1	1		
B	1		1	4	1	1	1		
C		1	5	4	1	1	1		
D	1		1	3	1	1	1		
E	1		1	3	4.5	0.5		1.5	0.5
F	1		1	3			4	1	1
G	1		1	3	0.5	4.5		0.5	1.5



Sum Contributions (+)

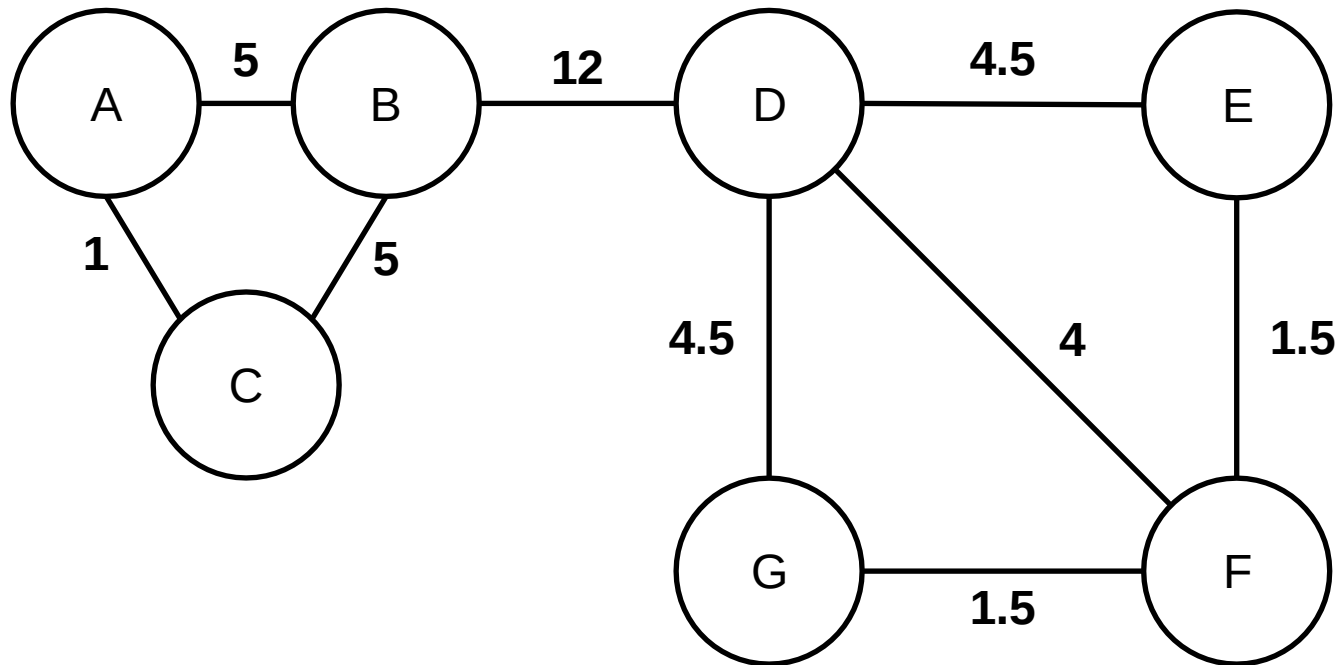
	AB	AC	BC	BD	DE	DG	DF	EF	GF
A	5	1		4	1	1	1		
B	1		1	4	1	1	1		
C		1	5	4	1	1	1		
D	1		1	3	1	1	1		
E	1		1	3	4.5	0.5		1.5	0.5
F	1		1	3			4	1	1
G	1		1	3	0.5	4.5		0.5	1.5
+	10	2	10	24	9	9	8	3	3

Sum Contributions (/2)

	AB	AC	BC	BD	DE	DG	DF	EF	GF
A	5	1		4	1	1	1		
B	1		1	4	1	1	1		
C		1	5	4	1	1	1		
D	1		1	3	1	1	1		
E	1		1	3	4.5	0.5		1.5	0.5
F	1		1	3			4	1	1
G	1		1	3	0.5	4.5		0.5	1.5
+	10	2	10	24	9	9	8	3	3
/2	5	1	5	12	4.5	4.5	4	1.5	1.5

Edge Contributions

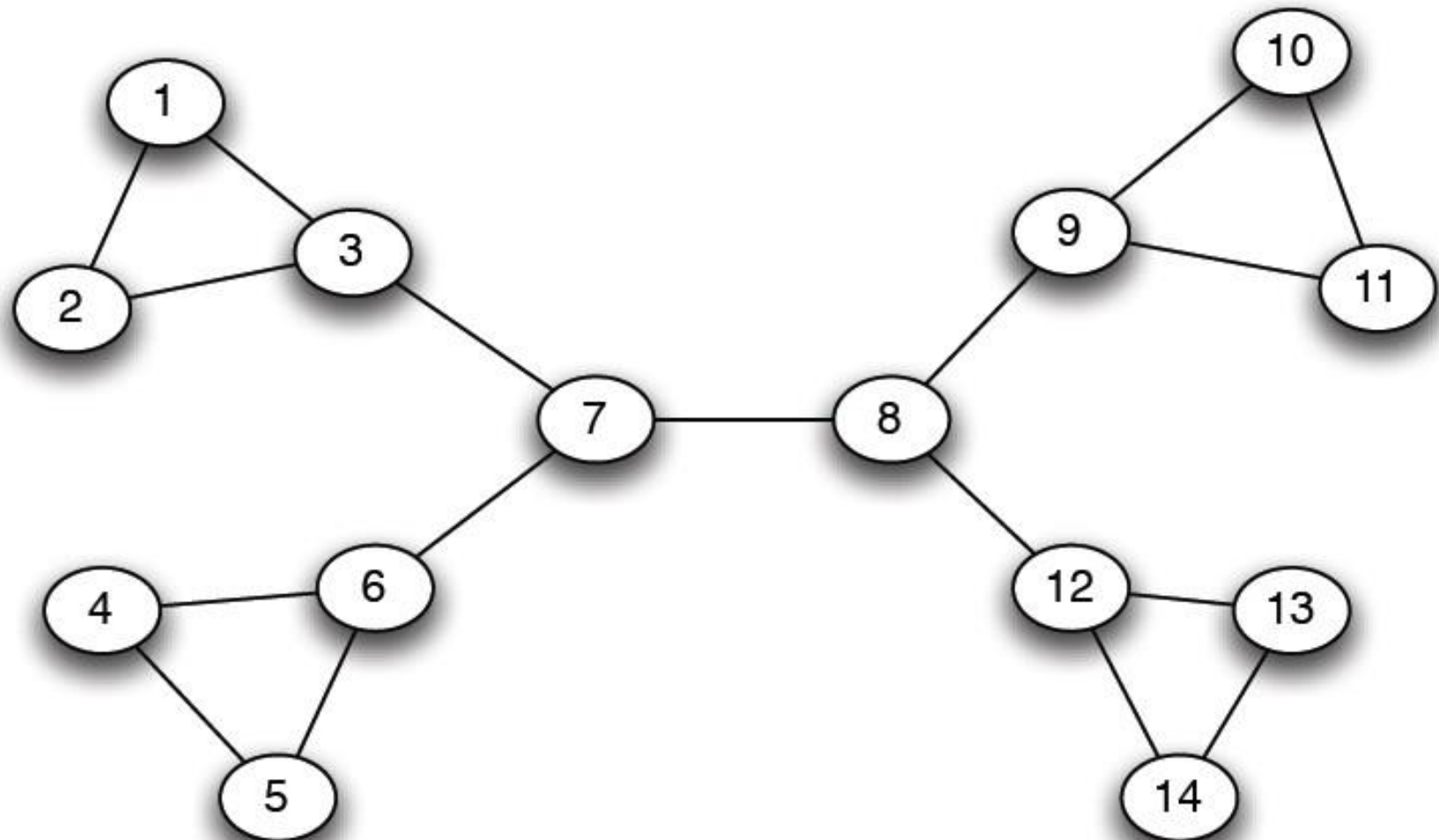
AB	AC	BC	BD	DE	DG	DF	EF	GF
5	1	5	12	4.5	4.5	4	1.5	1.5



Girvan-Newman

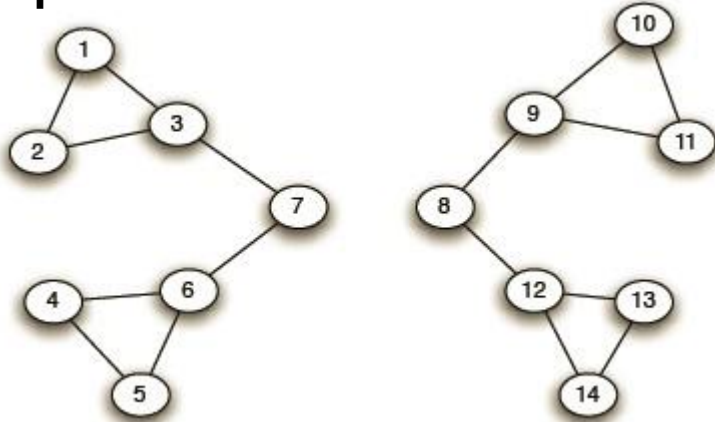
1. Repeat until no edges left
 - a) Calculate betweenness of edges
 - b) Remove edge(s) with highest betweenness

Example (0)

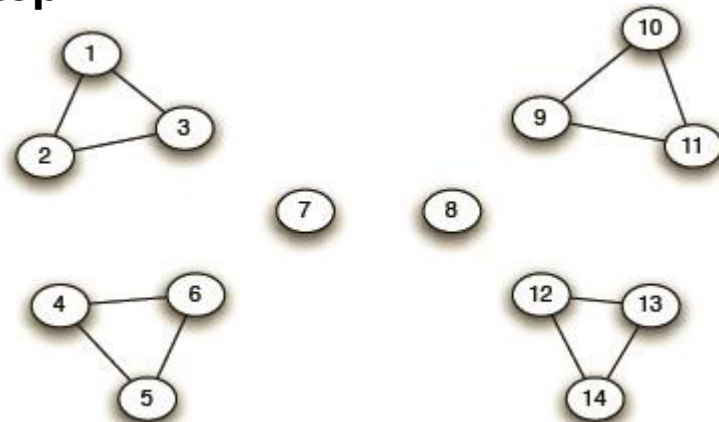


Example (1)

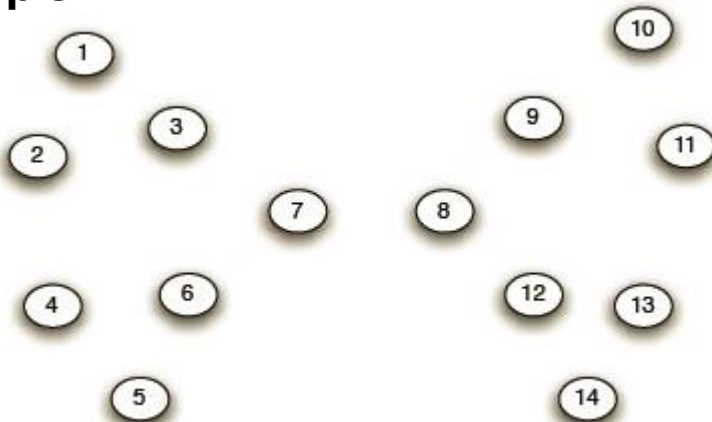
Step 1:



Step 2:



Step 3:



Hierarchical network decomposition:

